

Shu Quan

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Google Scholar

GitHub

Education

Peking University, Shenzhen Graduate School

Shenzhen, China

M.S. in Environmental Science, GPA: 3.61/4.0

Sep. 2024 – Jun. 2027 (Expected)

- Research focus: trustworthy AI, especially LLM/agent evaluation and safety; advisor: Prof. Yaodong Yang.
- Visiting research: LLM hallucination mitigation at HKUST(GZ) NLP-LAB; advisor: Prof. Xuming Hu.

Henan University

Kaifeng, China

B.S. in Geographic Information Science, GPA: 3.68/4.0, Rank: 1/46

Sep. 2020 – Jun. 2024

- Relevant coursework: Programming, Data Structures, Database Systems, Probability and Statistics.

Research Interests

Trustworthy AI; LLM/Agent Evaluation and Safety; Medical AI; AI for Science.

Publications and Preprints

Equal contribution is denoted by *.

1. **A Blind Spot in Alignment: Quantifying Biosecurity Risks in Large Language Models**
Shu Quan^{*}, Tianfang Hao^{*}, Sitong Fang, He Geng, Jiayi Zhou, Boyuan Chen, Kaile Wang, Donghai Hong, Juntao Dai, Yaodong Yang, Jiaming Ji
Under review, 2026
2. **SafeCLI: Exploring System-Centric Risks in CLI Agents via Augmented Observability**
Donghai Hong^{*}, Changhao Ju^{*}, Shu Quan^{*}, Yuyang Ren^{*}, Boyuan Chen, Jiayi Zhou, Kaile Wang, Juntao Dai, Yaodong Yang, Jiaming Ji
Under review, 2026
3. **Dense Clinical Contrasts Enhance Medical Knowledge Updating in Large Language Models**
Yangmin Huang^{*}, Shu Quan^{*}, He Geng, Xin Ye, Qianyun Du, Zhiyang He, Jiaxue Hu, Xiaodong Tao
Under review, 2026
4. **SLM-MATRIX: A Multi-Agent Trajectory Reasoning and Verification Framework for Enhancing Language Models in Materials Data Extraction**
Xin Li^{*}, Zhixuan Huang^{*}, Shu Quan, Cheng Peng, Xiaoming Ma
npj Computational Materials, 2025
5. **Nonlinear Effects of Blue-Green Space Variables on Urban Cold Islands in Zhengzhou Analyzed with Random Forest Regression**
Shu Quan^{*}, Maojuan Li^{*}, Tianqi Li, Haodong Liu, Yaohui Cui, Miaohan Liu
Frontiers in Ecology and Evolution, 2023
6. **Analysis of the PM_{2.5}–O₃ Pollution Characteristics and Its Potential Sources in Major Cities in the Central Plains Urban Agglomeration from 2014 to 2020**
Shu Quan, Miaohan Liu, Boxuan Chen, Yuehua Huang, Meijuan Wang, Qingxia Ma, Yan Han
Atmosphere, 2023

Research Experience

Peking University, PKU-Alignment Group
Research Assistant; advisor: Prof. Yaodong Yang

Sep. 2024 – Present
Beijing, China

- Built evaluation infrastructure for hidden failures in LLM and agent systems, including functional biosecurity risk (SPIKE-Bench), OS-level CLI-agent harms (SafeCLI), and layer-wise interpretability probes for physical properties in video VAE latents.
- Developed benchmarks, OS-level monitoring signals, and probing diagnostics that turn failures missed by surface metrics into observable evidence.

Alibaba Health
Research Intern, Tool-Use Evaluation for Medical LLM Agents

Apr. 2026 – Jun. 2026
Beijing, China

- Designed trajectory-aware diagnostics for medical LLM agents, decomposing tool use into tool selection, argument validity, evidence grounding, and pre-response verification across guideline lookup, drug-interaction checking, and triage routing; flagged under-calling, wrong-tool, invalid-argument, and ungrounded-evidence failures missed by final-answer accuracy.

OPPO AI Center, Large Model Algorithm Department
Research Intern

Jun. 2025 – Oct. 2025
Shenzhen, China

- Built evaluation protocols for DeepResearch long-form generation, covering planning quality, numerical claims, citation accuracy, source matching, and code-execution checks; factual accuracy improved from 60% to 78.5%.

iFLYTEK, South China Research Institute
Research Intern

Feb. 2025 – May 2025
Guangzhou, China

- Contributed to SEER-Bench by turning guideline updates and SEER oncology cases into temporally anchored evaluation and supervision data; under a same-budget SFT protocol, EMQ supervision helped Qwen3-4B reach 64.8%, with diagnostic gains linked to denser clinical-contrast signals and more economical representation updates.

Henan University
Undergraduate Researcher, Geospatial and Environmental Modeling

2022 – 2023
Kaifeng, China

- Conducted remote-sensing and statistical analyses of urban cold islands and atmospheric pollution, leading to first/co-first authored environmental science publications.

Honors and Awards

- Kaggle Silver Medal, LLM Adversarial Attack Competition, Rank 76/1,692 (Top 5%)
 - Outstanding Graduate, Henan Province
 - National Scholarship, China (Top 1%)
- Feb. 2025
Jun. 2024
Sep. 2023

Technical Skills

Programming: Python, PyTorch, Hugging Face Transformers, vLLM, LangGraph, Linux, Git
LLM methods: SFT, LoRA, reward modeling, GRPO, benchmark design, LLM-as-judge, multi-agent systems